

6

THE SERIES CIRCUIT

Circuits usually have several components that are connected in a particular way. This chapter explains the series circuit, in which each component in the circuit is connected one after the other. It describes how to apply Ohm's law to calculate current, voltage and resistance values in this type of circuit, and to determine the power taken by the whole circuit, and by each component in the circuit. We also look briefly at faults and their effects in a series circuit.

Chapter outline

- 6.1 The series circuit
- 6.2 Current in the series circuit
- 6.3 Resistance in the series circuit
- 6.4 Voltage in the series circuit
- 6.5 Summary of the series circuit
- 6.6 Power in the series circuit
- 6.7 Faults in the series circuit



6.1 The series circuit

The term 'connected in series' has been mentioned before. For example, in previous chapters we explained that an ammeter is connected in series to measure current, and cells are connected in series to make a battery. When components are connected in series, they are connected one after the other. Figure 6.1 shows four lamps and two cells, all connected in series.

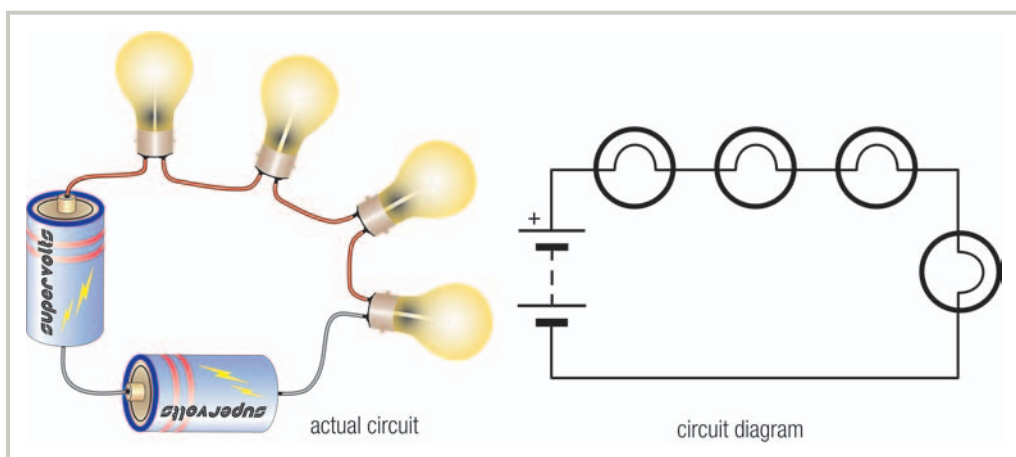


FIGURE 6.1

Lamps and cells connected in series

In a series circuit, as Figure 6.1 shows, each component has only *one* conductor connected to each terminal of the component. Because a component must have two terminals – one for the current to enter and the other for the current to flow out of the component – there can only ever be two conductors connected to each component in a series circuit.

6.2 Current in the series circuit

The circuit shown in Figure 6.2 has three resistors, (labelled R_1 , R_2 and R_3) connected in series with a battery. Notice that there is only *one* path for the current. In fact, a series circuit can be defined as a circuit in which there is only *one path for the current*.

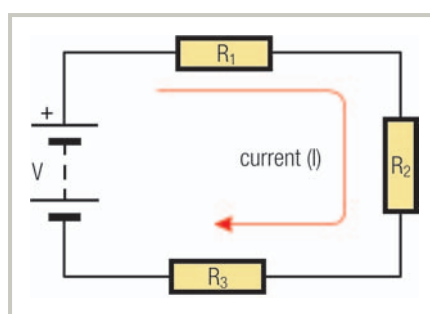


FIGURE 6.2

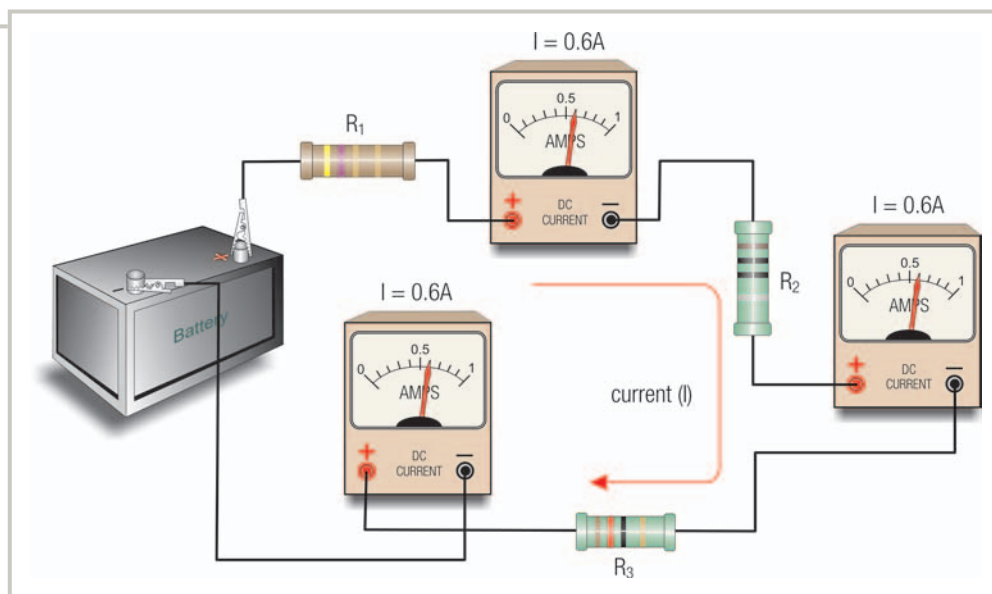
In a series circuit there is only one path for the current

Figure 6.3 shows a series circuit of three resistors, each of different values. An ammeter is connected after each resistor, to measure the current flowing in that resistor. The ammeters are also connected in series, so the whole circuit now has three resistors and three ammeters, all in series, with two cells also connected in series.

As you might expect, the ammeters all read the same value of current. In other words, *current is the same in all parts of a series circuit*. Therefore, it doesn't matter where an ammeter is connected in a series circuit, it will always read the same value of current.

FIGURE 6.3

Current is the same in all parts of a series circuit



SUMMING UP...

- A series circuit has only one path for the current.
- Current is the same in all parts of a series circuit.

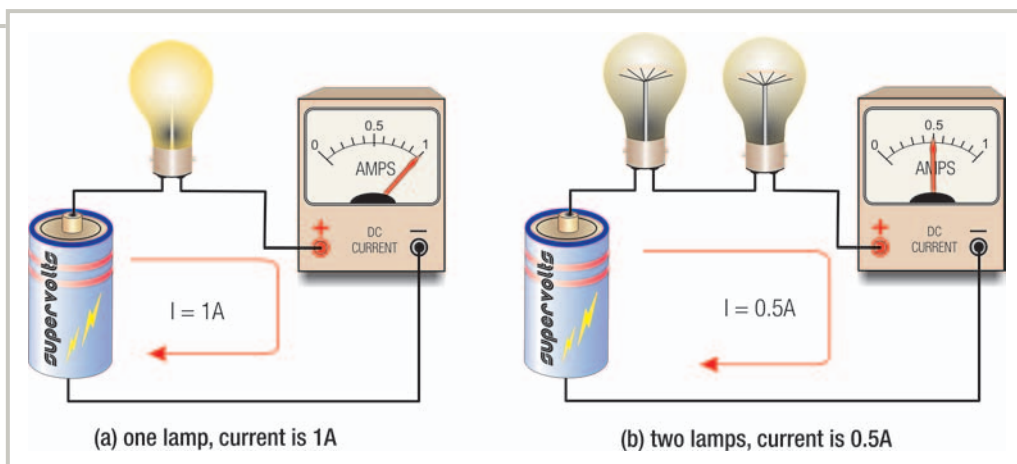
6.3 Resistance in the series circuit

In any circuit, the current depends on the resistance of the circuit and the value of the applied voltage. If two identical lamps are connected in series, there will be more resistance in the circuit than in a circuit with one lamp. Therefore the current will be less in the circuit that has two lamps.

This is shown in Figure 6.4, where the current is 1 A when one lamp is in the circuit. The lamp glows at its normal brightness. If a second lamp (with the same resistance as the first) is connected in series with the other lamp, the current falls to 0.5 A. Both lamps glow less brightly as the current is now half its previous value. Because the voltage hasn't changed, the resistance must have doubled.

FIGURE 6.4

Adding resistances in series increases the total circuit resistance



It follows that if more lamps are added in series, the current will drop to a lower value, and the lamps won't glow as brightly. In other words, connecting resistors in series *increases* the total resistance of the circuit. In fact, the total resistance (R_T) of the circuit can be found by adding the values of the individual resistors.

$$R_T = R_1 + R_2 + R_3 + \dots$$

where:

R_T is the total resistance of the series circuit in ohms

R_1, R_2, R_3 (etc.) are the individual resistance values in ohms.



Resistors in series

EXAMPLE 6.1

What is the total resistance between points A and B in Figure 6.5 below?

Solution

Values $R_1 = 120 \Omega$
 $R_2 = 330 \Omega$
 $R_3 = 820 \Omega$

Equation $R_T = R_1 + R_2 + R_3$
 $= 120 + 330 + 820$

Answer **$R_T = 1270 \Omega$**

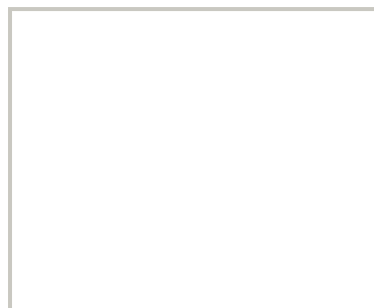


FIGURE 6.5

EXAMPLE 6.2

Find the total resistance between points A and B in Figure 6.6.

Solution

Values R_1, R_2 and $R_3 = 50 \Omega$

Equation $R_T = R_1 + R_2 + R_3$
 $= 50 + 50 + 50$

Answer **$R_T = 150 \Omega$**

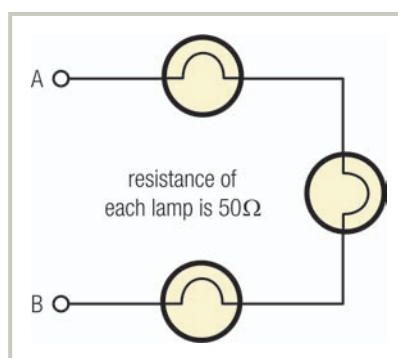


FIGURE 6.6

Example 6.2 shows that if all the resistance values in a series circuit are the same, the total resistance can be found by multiplying the resistance value by the number of resistors in the circuit. In Figure 6.6, there are three lamps, each with a resistance of 50Ω . The total resistance is therefore 3×50 , giving 150Ω .

Ohm's law

If you know the total resistance of a circuit and the applied voltage, you can find the current in the circuit with Ohm's law. To do this with a series circuit, you first find the total resistance (R_T), then use this value in the Ohm's law equation for current. The equations to use are:

$$R_T = R_1 + R_2 + R_3 \dots \text{(for total resistance)}$$

$$I = I_{R1} = I_{R2} = I_{R3} \dots \text{(current is the same in all components)}$$

$$I = \frac{V_T}{R_T} \text{ (to find the current).}$$

EXAMPLE 6.3

Calculate the current in Figure 6.5, if the applied voltage is 230 V.

Solution

Values $R_T = 1270 \, \Omega$ (as found in Example 6.1)
 $V_T = 230 \, \text{V}$

Equation $I = \frac{V_T}{R_T} = \frac{230}{1270}$

Answer **$I = 0.18 \, \text{A}$**

6.4 Voltage in the series circuit

The circuit in Figure 6.7 shows three resistors connected in series with a 20 V DC power supply. In this circuit:

- V_T is the applied voltage (the voltage of the voltage source) which causes a current to flow in the circuit
- V_1 , V_2 and V_3 are voltage drops. A voltage drop is caused by current flowing through a resistance. Its value is directly proportional to the value of the resistance (the higher the resistance the higher the voltage drop for the same value of current).

To calculate the voltage drop across each resistor, first find the current (I) in the circuit, then find the voltage drop across each resistor with Ohm's law. Example 6.4 explains.

EXAMPLE 6.4

Calculate the voltage drops across the three resistors in Figure 6.7.

Solution

Values $V_T = 20 \, \text{V}$
 $R_1 = 20 \, \Omega$, $R_2 = 30 \, \Omega$, $R_3 = 50 \, \Omega$
 V_1 , V_2 , $V_3 = ?$

Equations $I = \frac{V_T}{R_T}$
 $R_T = R_1 + R_2 + R_3$
 $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$

Working $R_T = 20 + 30 + 50 = 100 \, \Omega$
 $I = \frac{20}{100} = 0.2 \, \text{A}$

Answers $V_1 = 0.2 \times 20 = 4 \, \text{V}$
 $V_2 = 0.2 \times 30 = 6 \, \text{V}$
 $V_3 = 0.2 \times 50 = 10 \, \text{V}$



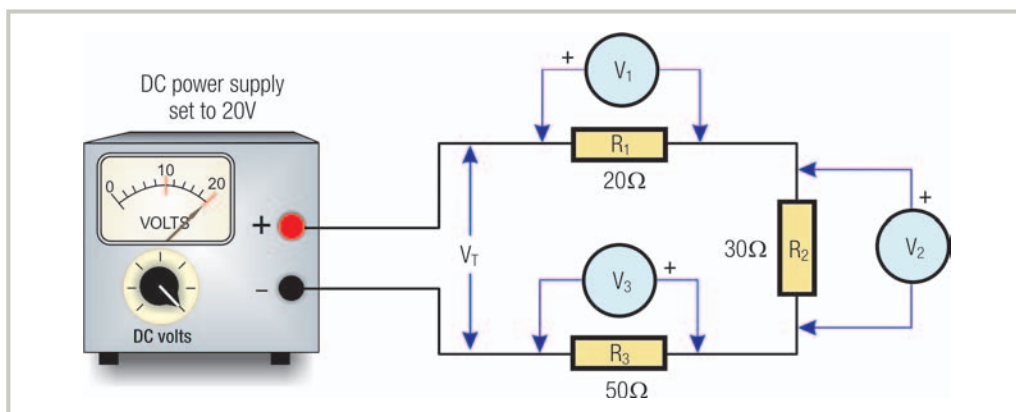


FIGURE 6.7

Kirchhoff's voltage law

As you would expect, when you add up the voltage drops in Example 6.4, you get 20 V, which is the supply voltage. This is true for all series circuits, where *the sum of the voltage drops equals the supply voltage*. This is called Kirchhoff's voltage law (after Gustav Kirchhoff, a German physicist), which says:

The algebraic sum of the voltage drops in a series circuit equals the applied voltage.

This law is as important as Ohm's law. Like Ohm's law, it can never be wrong. The voltage drops around a series circuit can *never* total any more, or any less, than the applied voltage. The next two examples show how to use Kirchhoff's voltage law.

 Kirchhoff's voltage law

EXAMPLE 6.5

What is the value of the supply voltage (V_T) in the circuit shown in Figure 6.8?

Solution

Values $V_1 = 11\text{ V}$
 $V_2 = 23\text{ V}$
 $V_3 = 6\text{ V}$
 $V_T = ?$

Equation $V_T = V_1 + V_2 + V_3$
 $= 11 + 23 + 6$

Answer $V_T = 40\text{ V}$

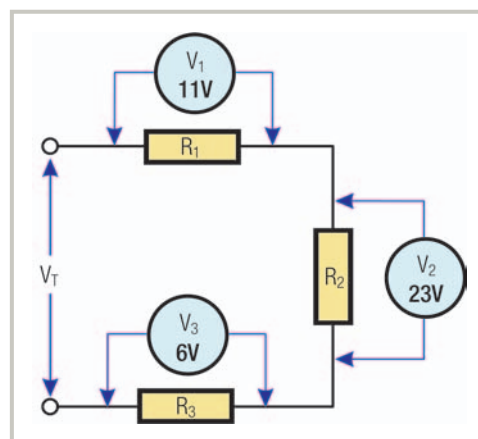
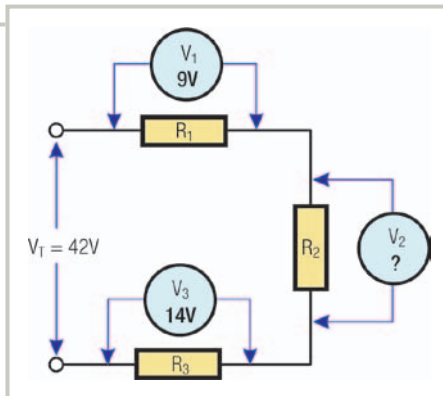


FIGURE 6.8

If the supply voltage and all but one of the voltage drops in a series circuit are known, you can calculate the unknown voltage drop by adding the known voltage drops and subtracting their sum from the value of the supply voltage.

EXAMPLE 6.6

Calculate the voltage drop across resistor R_2 in the circuit of Figure 6.9.

FIGURE 6.9**Solution**

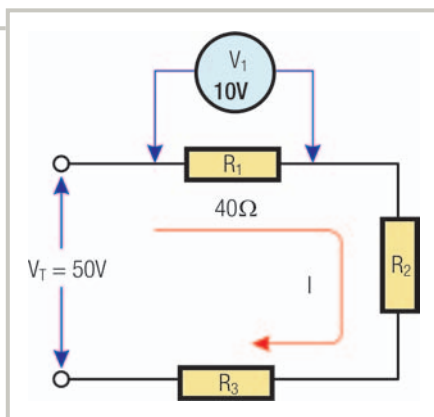
Values	$V_1 = 9\text{ V}$
	$V_2 = ?$
	$V_3 = 14\text{ V}$
	$V_T = 42\text{ V}$
Equation	$V_T = V_1 + V_2 + V_3$
transposed	$V_2 = V_T - (V_1 + V_3)$
	$= 42 - (9 + 14)$
Answer	$V_2 = 19\text{ V}$

Ohm's law and Kirchhoff's voltage law

If you know the voltage drop across a known value of resistance in a series circuit, you can use Ohm's law to find the current flowing in this resistor, and therefore the whole circuit, because the current in a series circuit is the same in all parts. Knowing the current lets you calculate the total resistance of the circuit, assuming you know the supply voltage. The following examples demonstrate this.

EXAMPLE 6.7

Find the total resistance of the circuit in Figure 6.10.

FIGURE 6.10**Solution**

Values	$V_T = 50\text{ V}$
	$R_1 = 40\ \Omega$
	$V_1 = 10\text{ V}$
	$R_T = ?$
Equations	$R_T = \frac{V_T}{I}$
	$I = \frac{V_1}{R_1}$
Working	$I = \frac{10}{40} = 0.25\text{ A}$
	$R_T = \frac{50}{0.25}$
Answer	$R_T = 200\ \Omega$

The values of R_2 and R_3 in Figure 6.10 can also be found, providing you know the voltage drop across either one of these resistors. Example 6.8 shows how to apply Kirchhoff's voltage law and Ohm's law to find these values. It assumes the same values of current and total resistance from Example 6.7.

EXAMPLE 6.8

If the total resistance of the circuit in Figure 6.11 is $200\ \Omega$, what are the values of R_2 and R_3 ? Also, what's the voltage (V_2) across R_2 ?

Solution

Values $R_T = 200\ \Omega$, $R_1 = 40\ \Omega$
 $I = 0.25\ \text{A}$ (as found in Example 6.7)

$$V_T = 50\ \text{V}$$

$$V_1 = 10\ \text{V}, V_3 = 25\ \text{V}$$

Equations $R_3 = \frac{V_3}{I}$

$$R_2 = R_T - (R_1 + R_3)$$

$$V_2 = V_T - (V_1 + V_3)$$

Working $R_3 = \frac{25}{0.25}$

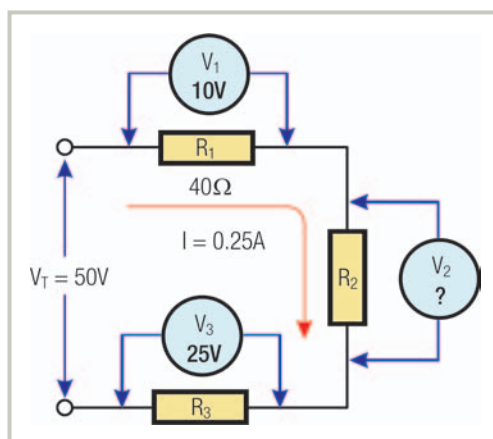
Answer **$R_3 = 100\ \Omega$**

$$R_2 = 200 - (40 + 100)$$

Answer **$R_2 = 60\ \Omega$**

$$V_2 = 50 - (10 + 25)$$

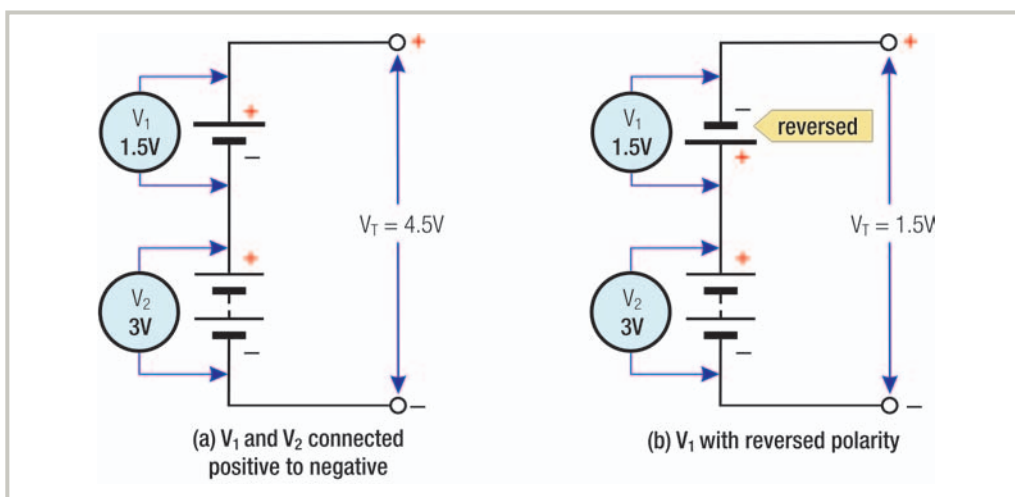
Answer **$V_2 = 15\ \text{V}$** (V_2 also equals $IR_2 = 0.25 \times 60 = 15\ \text{V}$)

**FIGURE 6.11**

Voltage sources

Kirchhoff's voltage law says that voltages are added *algebraically*, which means you have to consider the *polarity* of the voltage. For example, as shown in Figure 6.12(a), when cells are connected in series, they are arranged so the positive terminal of a cell connects to the negative terminal of the next cell. The total voltage is the sum of the individual cell voltages.

Figure 6.12(b) shows the effect if a cell is connected in reverse to the others. As expected, the total voltage is reduced, as the voltage of the reversed cell is subtracted from the other cell voltages.

**FIGURE 6.12**

Voltages are added algebraically, so a reversed cell will cause the applied voltage to be reduced

6.5 Summary of the series circuit

As we've shown, current, voltage and resistance values in a series circuit can be calculated in various ways, depending on what you know about the circuit. The following summarises how to calculate resistance, current and voltage in a series circuit.

Resistance

Total resistance of a series circuit



The total resistance of a series circuit *always* equals the sum of the individual resistance values.

$$1 \quad R_T = \text{sum of each resistance value. That is, } R_T = R_1 + R_2 + R_3 + \dots$$

If all the resistance values are the same, the total resistance of the circuit equals the number of resistors in the circuit multiplied by the value of one resistor.

$$2 \quad R_T = R \times n \text{ (where } n = \text{number of resistors in the circuit and } R = \text{value of one resistor).}$$

Ohm's law is used to find the total resistance if the applied voltage and the circuit current are known. That is,

$$3 \quad R_T = \frac{V_T}{I}.$$

There are two ways to find the value of *one* resistor in a series circuit. The first is to transpose equation 1 in terms of the unknown resistor (R_x). This equation assumes you know all but one of the resistance values in the circuit.

$$4 \quad R_x = R_T - (\text{sum of all the other resistance values})$$

The second way is with Ohm's law, where:

$$5 \quad R_1 = \frac{V_1}{I}, R_2 = \frac{V_2}{I} \text{ and so on.}$$

Current

Current is the same in all parts of a series circuit



The current in a series circuit is always *the same* in all parts of the circuit. Current is found with Ohm's law as shown in the next two equations. Use equation 6 if you know the applied voltage and the total resistance of the circuit.

$$6 \quad I = \frac{V_T}{R_T}$$

If you know the voltage across any *one* resistance in the circuit and you also know the value of that resistance, you can use equation 7 to find the current.

$$7 \quad I = \frac{V_1}{R_1} = \frac{V_2}{R_2} \text{ and so on}$$

To measure circuit current, connect an ammeter in *any part* of a series circuit. If you know the current in one component, you then know the current in all others in the circuit.

Voltage

The sum of the voltage drops in a series circuit equals the applied voltage



In a series circuit there is an applied voltage (called V_T for total voltage) and a number of voltage drops (V_1 , V_2 and so on). A voltage drop is caused by current flowing in a resistance – the applied voltage causes the current in the first place. By Kirchhoff's voltage law, the sum of the voltage drops in a series circuit *always* equals the applied voltage. Use Kirchhoff's voltage law or Ohm's law

to find either the applied voltage or a voltage drop, depending on what you know about the circuit. Use equations 8 or 9 to find the applied voltage (V_T).

$$8 \quad V_T = V_1 + V_2 + V_3 + \dots \text{ (Kirchhoff's voltage law)}$$

$$9 \quad V_T = I \times R_T \text{ (Ohm's law)}$$

To find a voltage drop (V_x) with Kirchhoff's voltage law, equation 8 has to be transposed in terms of the unknown voltage. That is:

$$10 \quad V_x = V_T - (\text{sum of all the other voltage drops in the circuit})$$

Ohm's law can also be used, where:

$$11 \quad V_1 = I \times R_1, V_2 = I \times R_2 \text{ and so on}$$

For equation 11, as the current is the same in all parts of the circuit, it follows that the higher the resistance, the higher the voltage drop. That is, the voltage drop across a resistor in a series circuit is *directly proportional* to its resistance.

The highest voltage drop in a series circuit is across the highest resistance

EXERCISE 6.1

- 1 In Figure 6.13 below, for circuit (a) find current I , for circuit (b) find resistance R_2 and for circuit (c) find the supply voltage V_T .

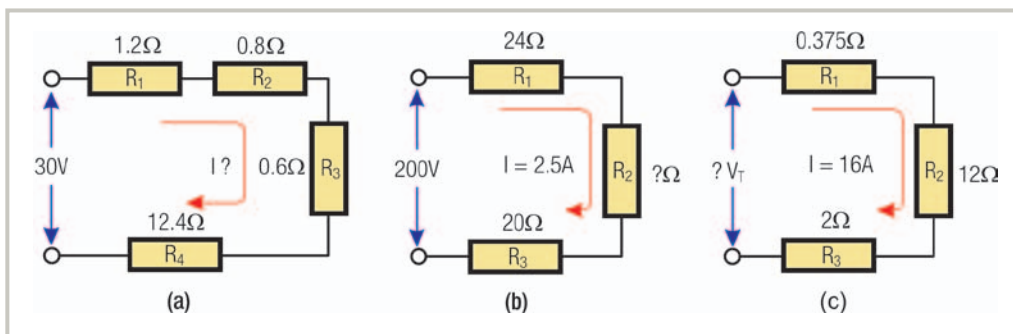


FIGURE 6.13

- 2 In Figure 6.14(a) find current I and supply voltage V_T .
- 3 In Figure 6.14(b) find the supply voltage V_T , current I and the values of resistances R_2 , R_3 and R_4 .

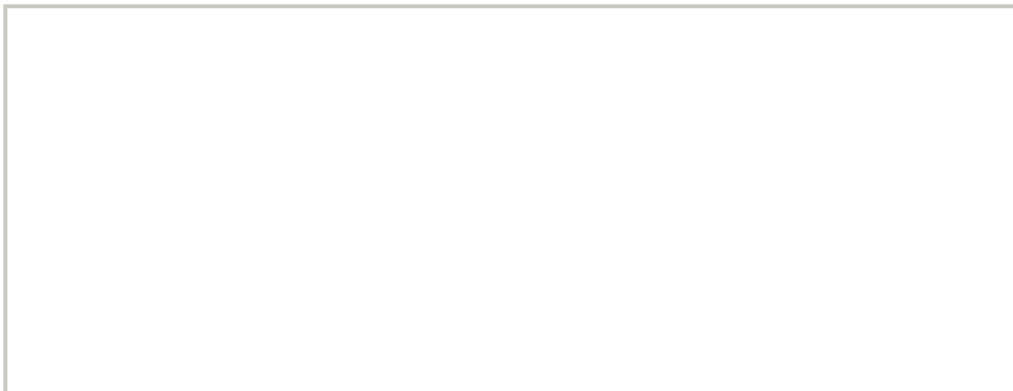


FIGURE 6.14

6.6 Power in the series circuit

The total power, or the power taken by an individual component in a series circuit, can be found with the power equations explained in Chapter 4. Ohm's law and the equations introduced in this chapter are also often needed.

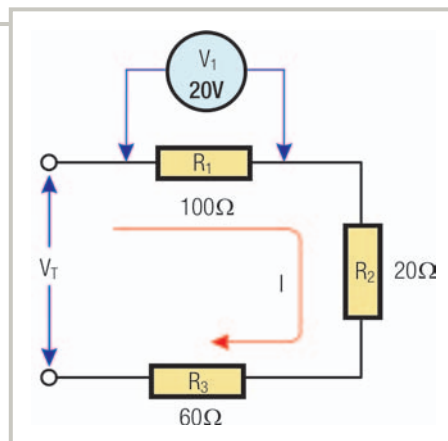
Calculating total power

If the circuit current (I) and the supply voltage (V) are known, the total power (P_T) taken by a circuit is the product of the current I and the voltage V . That is, $P = VI$ watts. This applies to all types of circuits. Another way, if you know the current and resistance of the circuit, is with the equation $P_T = I^2 R_T$. As Example 6.9 shows, sometimes you also have to find the circuit current and total resistance from values associated with one component in the circuit.

EXAMPLE 6.9

Find the total power taken by the circuit in Figure 6.15.

FIGURE 6.15



Solution

Values	$V_1 = 20\text{ V}$ $R_1 = 100\ \Omega$ $R_2 = 20\ \Omega$ $R_3 = 60\ \Omega$ $P_T = ?$
Equations	$P_T = I^2 R_T$ $I = \frac{V_1}{R_1}$ $R_T = R_1 + R_2 + R_3$
Working	$I = \frac{20}{100} = 0.2\text{ A}$ $R_T = 100 + 20 + 60$ $R_T = 180\ \Omega$ $P_T = 0.2^2 \times 180$
Answer	$P_T = 7.2\text{ W}$

Calculating power taken by individual components

Example 6.10 shows how to find the power taken by each resistor in Figure 6.15. Figure 6.16 is the same as before, except it shows the value of the current, which was found to be 0.2 A.

EXAMPLE 6.10

Calculate the power taken by each resistor in the circuit of Figure 6.16.

Solution

Values $V_1 = 20\text{ V}$
 $R_1 = 100\ \Omega$
 $R_2 = 20\ \Omega$
 $R_3 = 60\ \Omega$
 $I = 0.2\text{ A}$ (from Example 6.9)
 $P_1, P_2, P_3 = ?$

Working $P_1 = \frac{V_1^2}{R_1} = \frac{20^2}{100}$

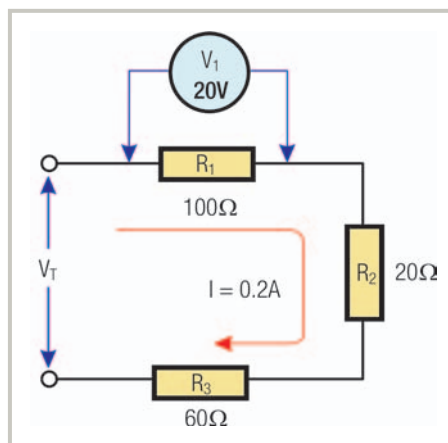
Answer **$P_1 = 4\text{ W}$**

Working $P_2 = I^2 R_2 = 0.2^2 \times 20$

Answer **$P_2 = 0.8\text{ W}$**

Working $P_3 = I^2 R_3 = 0.2^2 \times 60$

Answer **$P_3 = 2.4\text{ W}$**

**FIGURE 6.16**

In this example, the power taken by R_1 was calculated using a different equation to the one used to find the power taken by the other two resistors, but we could have used the equation $P_1 = I^2 R_1$. As a check, $P_1 = 0.2^2 \times 100$, which gives the same answer of 4 W.

Pulling it all together

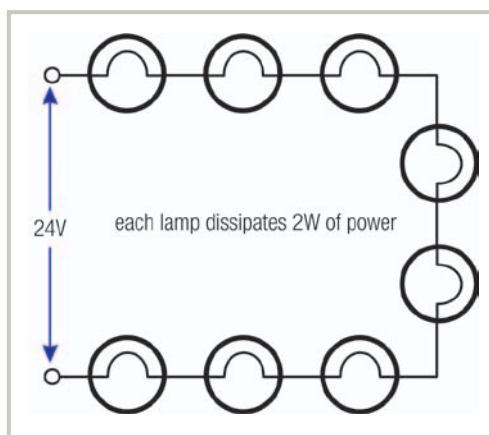
Adding the individual power values found in Example 6.10 gives, as you would expect, the total power calculated in Example 6.9. Therefore, in a series circuit:

$$P_T = P_1 + P_2 + P_3 + \dots$$

If you know the total power and the power taken by every component except one, the unknown power consumption can be found by transposing this equation in terms of the unknown power (P). That is,

$$P = P_T - (\text{sum of the remaining individual values of power}).$$

Example 6.11 uses a number of equations to find all the unknowns in the series circuit shown in Figure 6.17, which is a string of eight 2 W lamps.

**FIGURE 6.17****EXAMPLE 6.11**

For the circuit in Figure 6.17, calculate the:

- 1 total power taken by the circuit
- 2 circuit current
- 3 equivalent resistance of the circuit
- 4 resistance of each lamp
- 5 voltage across each lamp.





Solution

Values $V_T = 24 \text{ V}$
 $P_{\text{lamp}} = 2 \text{ W}$
 number of lamps (n) = 8

$$1 \quad P_T = P_{\text{lamp}} \times n = 2 \times 8$$

$$\mathbf{P_T = 16 \text{ W}}$$

$$2 \quad I = \frac{P_T}{V_T} = \frac{16}{24}$$

$$\mathbf{I = 0.67 \text{ A}}$$

$$3 \quad R_T = \frac{V_T}{I} = \frac{24}{0.67}$$

$$\mathbf{R_T = 36 \Omega}$$

$$4 \quad R_{\text{lamp}} = \frac{R_T}{n} = \frac{36}{8}$$

$$\mathbf{R_{\text{lamp}} = 4.5 \Omega}$$

$$5 \quad V_{\text{lamp}} = \frac{V_T}{n} = \frac{24}{8} \text{ or } V_{\text{lamp}} = I \times R_{\text{lamp}} = 0.67 \times 4.5$$

$$\mathbf{V_{\text{lamp}} = 3 \text{ V}}$$

6.7 Faults in the series circuit

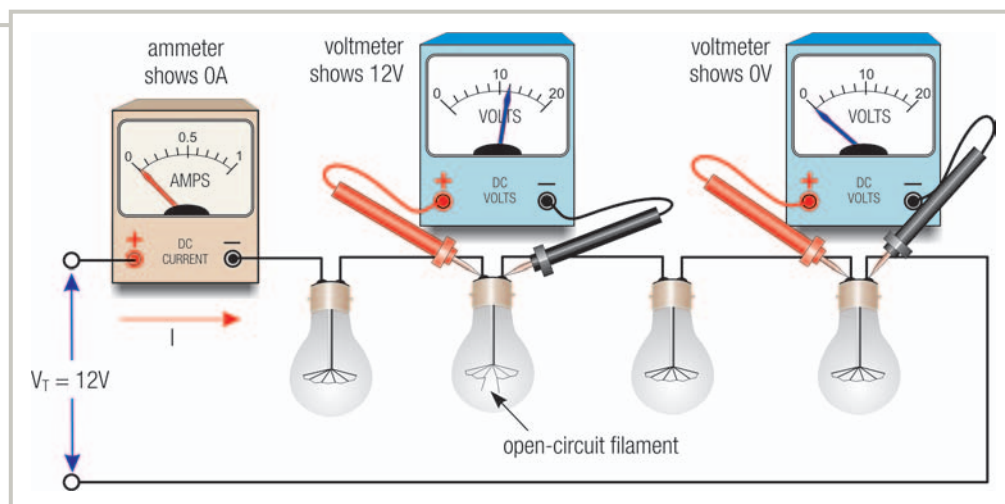
The two most common faults in any circuit are an open-circuit or a short-circuit. A third type of fault is a change in the value of a resistance within the circuit. We look now at the effects these faults have in a series circuit.

Open-circuit

The effects of an open-circuit are shown in Figure 6.18. In this diagram, one lamp has a blown filament, and is therefore an open-circuit. As a result, the current is zero, the voltage across the open-circuit equals the supply voltage and the voltage across all other lamps is zero.

FIGURE 6.18

Effects of an open-circuit in a series circuit



Therefore, when there's an open-circuit in any part of a series circuit:

- the resistance of the circuit is infinity (open-circuit)
- the current (and therefore the power taken) is zero
- the voltage across the open-circuit equals the applied voltage (a useful fault-finding hint)
- the voltage across all other components equals zero
- all components in the circuit stop working.

Short-circuit

A short-circuit across a component in a series circuit will *always* cause the current to increase, because the total resistance of the circuit drops.

In Figure 6.19, one lamp has an internal short-circuit. If the lamps each have a resistance of $3\ \Omega$, the total resistance of the circuit (without the short-circuit) is $12\ \Omega$. By Ohm's law, the current is $1\ \text{A}$ ($I = 12\ \text{V}/12\ \Omega$). The voltage across each lamp is $3\ \text{V}$ ($V = 1\ \text{A} \times 3\ \Omega$).

The short-circuit will cause the total resistance of the circuit to drop by $3\ \Omega$ (from $12\ \Omega$ to $9\ \Omega$) and the current is now $1.33\ \text{A}$ ($12\ \text{V}/9\ \Omega$). The voltage across the lamp with the short-circuit is zero, as the resistance of the short-circuit is virtually zero. The voltage drop across the other three lamps is now $4\ \text{V}$ ($1.33\ \text{A} \times 3\ \Omega$), instead of $3\ \text{V}$. These lamps will glow more brightly because of the increased voltage across them.

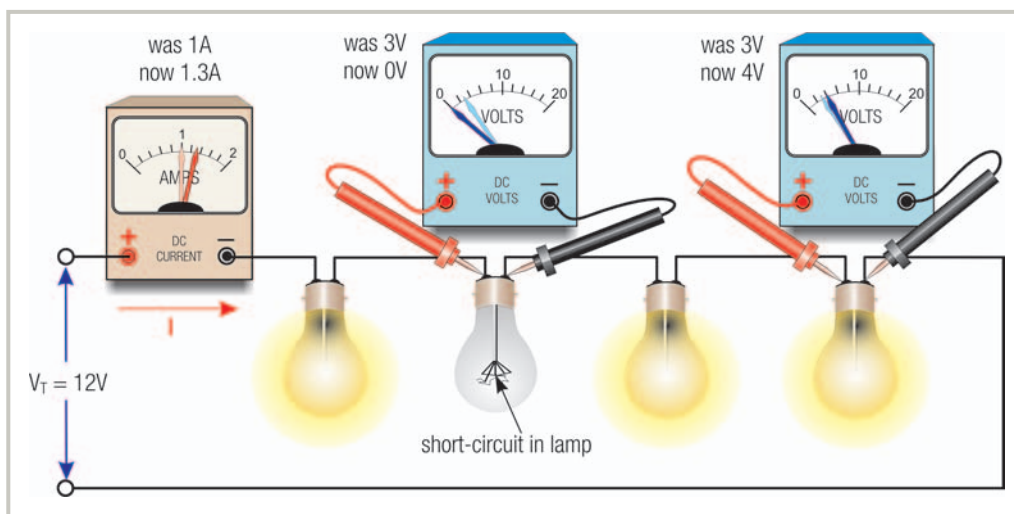


FIGURE 6.19

A short-circuit in a series circuit always causes the current to increase

Therefore, when there's a short-circuit across a component in a series circuit the:

- circuit resistance drops
- circuit current (and therefore the power taken) increases
- voltage drop across the remaining components increases
- voltage drop across the short-circuit equals zero
- components in the rest of the circuit work harder because of the increased voltage drop across them.

The effect on the circuit depends on how much the current increases. In Figure 6.19 the lamps are operating at a higher than normal voltage, which might cause them to burn out (become open-circuit).

Changed resistance

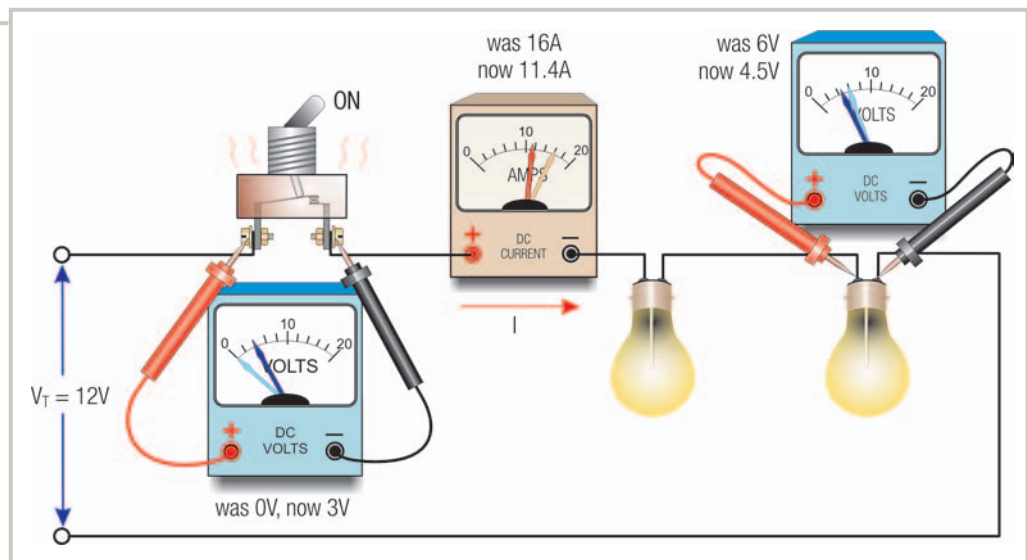
All components in a series circuit have a certain value of resistance. If the resistance of one of these components changes, perhaps due to an internal fault, ageing or external factors, the total resistance of the circuit changes. The effect depends on whether the resistance has increased or decreased, and by how much. The extremes are an open-circuit or a short-circuit, as explained above.

An example is a switch with dirty contacts. Normally, the contacts should have virtually no resistance when the switch is turned on, so a switch is therefore not usually considered to be a resistance. However, if the contacts are oxidised, worn or are not touching each other properly, the switch becomes a measurable resistance that affects the operation of the circuit.

Figure 6.20 shows a 12 V DC circuit with a switch and two lamps connected in series. When the circuit is working correctly, 16 A of current flows, and both lamps have 6 V across them. However, a faulty switch has caused the circuit current to fall to 11.4 A, and the voltage across the lamps is now 4.5 V. As a result, they are not glowing as brightly as usual. Also, the switch will heat up due to the power being dissipated by the resistive contacts.

FIGURE 6.20

A faulty switch with an increased resistance between its contacts affects the rest of the circuit



When the fault is an *increased* resistance in a series circuit the:

- circuit resistance increases
- circuit current (and therefore the power taken) decreases
- voltage drop across the remaining components decreases
- voltage drop across the increased resistance increases
- components in the rest of the circuit receive less power.

When the fault is a *decreased* resistance in a series circuit the:

- circuit resistance decreases
- circuit current (and therefore the power taken) increases
- voltage drop across the remaining components increases
- voltage drop across the reduced resistance decreases
- components in the rest of the circuit receive more power.

CHAPTER SUMMARY

In a series circuit:

- current is the same in all parts
- total resistance (R_T) is the sum of the individual resistances
- the algebraic sum of the voltage drops in the circuit equals the supply voltage V_T (Kirchhoff's voltage law)
- Ohm's law is used to find unknown voltage, resistance and current values in the circuit
- total power (P_T) taken equals the supply voltage times the current ($P_T = V_T I$)
- total power can also be found with the equations $P_T = I^2 R_T$ or $P_T = \frac{V_T^2}{R_T}$
- the sum of the power taken by each component equals the total power
- an open-circuit anywhere in the circuit prevents any component from receiving power, as the circuit current is zero
- the voltage drop across an open-circuit equals the supply voltage
- a short-circuit will cause an increase in the circuit current
- the voltage drop across a short-circuit equals zero
- effects caused by faults that cause a change in resistance will be somewhere between the extremes of an open- or a short-circuit.

SELF-CHECK

Answer these questions then check your answers at the back of the book.

- 1 The current in a component in a series circuit measures 25 A. What is the total current in the circuit?
- 2 A circuit has three resistances, each of 50 ohms, connected in series. What is the total resistance of the circuit?
- 3 In the circuit of Question 2, the voltage across one resistor is 15 V. Find the total voltage applied to the circuit.
- 4 In a series circuit, which component has the highest voltage drop across it?
- 5 A series circuit with four components (R_1 to R_4) has a total resistance of 800 Ω . If the resistance of R_1 is 25 ohms, R_2 is 75 ohms and R_3 is 500 ohms, what is the resistance of R_4 ?
- 6 For the circuit in Figure 6.21, calculate:
 - a the total resistance
 - b the voltage drops across all four resistors (R_1 to R_4)
 - c total power taken by the circuit
 - d power taken by R_3

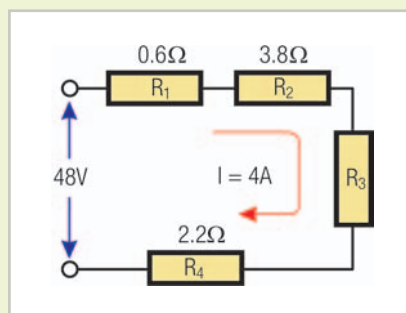


FIGURE 6.21

- 7 The circuit in Figure 6.22 has three identical lamps, each dissipating 150 W of power. Find:
 - a the voltage drop across each lamp
 - b the resistance of each lamp
 - c total resistance of the circuit
 - d current flowing in the circuit

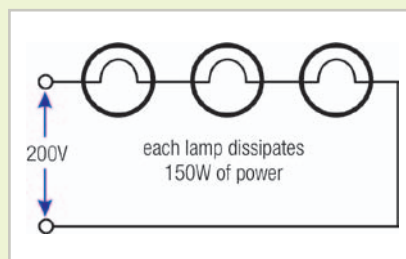
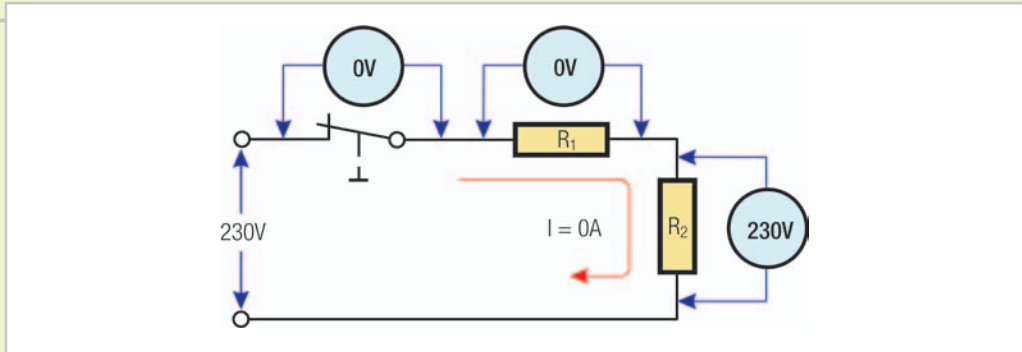


FIGURE 6.22

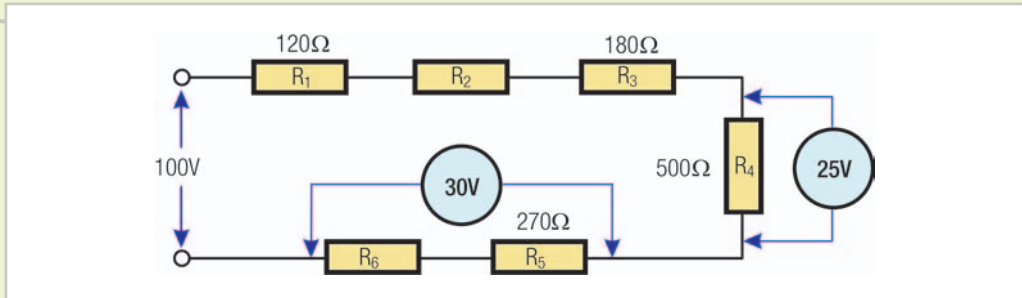
- 8 Which component in Figure 6.23 is faulty, what is the fault and how did you arrive at your conclusion?
- 9 What would be the voltage readings in Figure 6.23 if resistor R_1 is a short-circuit, and resistor R_2 is functioning normally?

FIGURE 6.23



- 10 For the circuit in Figure 6.24, find the:
- a circuit current
 - b resistance of resistor R_6
 - c total resistance of the circuit
 - d resistance of resistor R_2

FIGURE 6.24



ONLINE RESOURCES

For more information about the contents of this chapter, the best search term is 'series circuit'. Look for websites with a .edu in the address, as these usually give uncomplicated descriptions.